

Documentation of the project

Purpose:

This is a **DSA-based quiz system** in C++ that allows a student to:

- First program ask the student how many students want to take quiz then
 - Enter their name and roll number
 - Select a quiz subject (**C++, Java, or HTML**)
 - Attempt a multiple-choice quiz
 - Validate answers (**accept only A, B, C, D**)
 - To store each student's result
 - To search student record there are options
 - Search student record by name or roll no
 - Display all students record
 - Display result record according to highest marks
 - Display a result with marks, percentage, and pass/fail status
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Key concepts covered:

- Linked list (Dynamic data structure)
 - Function and modular programming
 - Recursion
 - Sorting algorithm
 - Input validation
 - Condition and looping logic
 - pointers
 - Basic data handling
 - System command (clear screen)
 - Dynamic memory location
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1. linked list (dynamic data structure):

- The program uses a struct Student containing name, subject, score, and a next pointer.
- This implements a **singly linked list**, a fundamental data structure in computer science.
- **Usage in program:**

- **Insertrecord()** – inserts new student nodes at the end of the list.
 - **Displayall()** – traverses the list to display student results.
 - **searchrec()** – recursively searches the list for a student by name or roll no.
 - **sortByscore()** – sorts nodes by score in descending order.
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2. Function and modular programming:

Each operation is encapsulated in its own function:

- **getValidChoice()** – input validation.
 - **takeQuiz()** – handles quiz logic and scoring.
 - **insertRecord()** – inserts nodes into the linked list.
 - **displayAll()** – prints all student records.
 - **searchRec()** – recursive search function.
 - **sortByScore()** – sorts the linked list by score.
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3. Recursion:

Searching of a specific student record by name is done using a **recursive function**

- **searchRec(Student *head, string target)** recursively traverses the linked list to find a student.
 - **Base case:** when the list is empty (head == NULL) or the target is found.
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4 .Sorting algorithm:

- Records are sorted by score using nested loops (Bubble Sort logic)
 - **sortByScore()** sorts students by score in descending order using nested loops.
 - **swap()** is used to exchange node data.
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5. input validation:

- **getValidChoice()** ensures that only valid choices (**A/B/C/D**) are accepted.
- If users enter other than A/B/C/D program gives **invalid choice please try again**.

6 .Condition and looping logic:

- **Switch statement** handles subject selection for the quiz.
- **for loops** for iterating over students during input.
- **while** loops for traversing the linked list and validating input.

7 .Pointers:

- **Student *head** and **Student *ptr** manage the linked list.
- **next** pointers create the chain of nodes.

5. Program Flow in main()

1. Ask how many students want to take quiz
2. Clears screen
3. Takes student name and roll number
4. In a loop:
5. Asks for subject selection
6. Based on selection:
 - Creates an object of respective quiz function (dynamically)
 - Calls **takequiz()** using pointer
 - Displays quiz result
 - Asks if user wants to attempt another quiz
 - Deletes the quiz object
7. Exits if user chooses not to attempt another quiz
8. Display result for all students

6 .Result display:

- The **displayAll()** function traverses the linked list of student records and prints all students' quiz results in a **formatted, human-readable list**.
 - It is used to **provide a complete overview of all stored quiz records**.
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Example output:

=====Quiz management system=====

How many students want to attempt quiz? 2

Enter student 1 name: Zainab Nadeem

Enter roll no: 1

Select Quiz Subject:

1. C++

2. Java

3. HTML

Enter choice: 1

----- C++ QUIZ -----

1. Which is OOP concept?

A. Polymorphism

B. Macro

C. Pointer

D. Structure

Enter choice (A/B/C/D): A

Your answer is correct!

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Zainab , Your Score = 5 /5

Enter Student 2 Name: sara

Enter roll no: 2

Select Quiz Subject:

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===== RESULT MENU =====

1. Show All Results

2. Search Result (Recursion)

3. Sort Result by Score

4. Exit

Enter choice: 2

Enter student name to search: Zaianb

Record Found!

Name: Zainab

Subject: c++

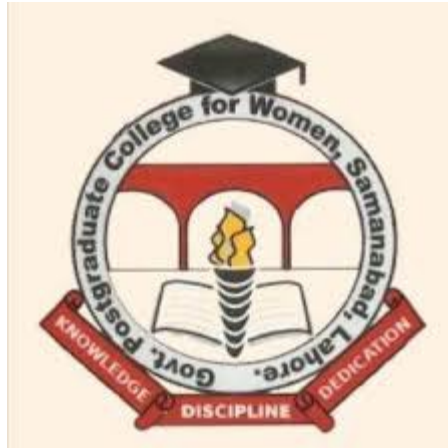
Score: 5/5

Percentage: 100.0%

Result: pass

Summary of Concepts Covered

- Linked List (Insertion, Traversal, Search, Sorting)
 - Functions & Modular Programming
 - Input Validation
 - Conditional Statements (if-else, switch)
 - Recursion
 - Sorting (Nested Loops / Bubble Sort)
 - Loops (for and while)
 - Pointers & Dynamic Memory Management
 - Strings (std::string)
 - System Commands (CLS)
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Quiz Base Project

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